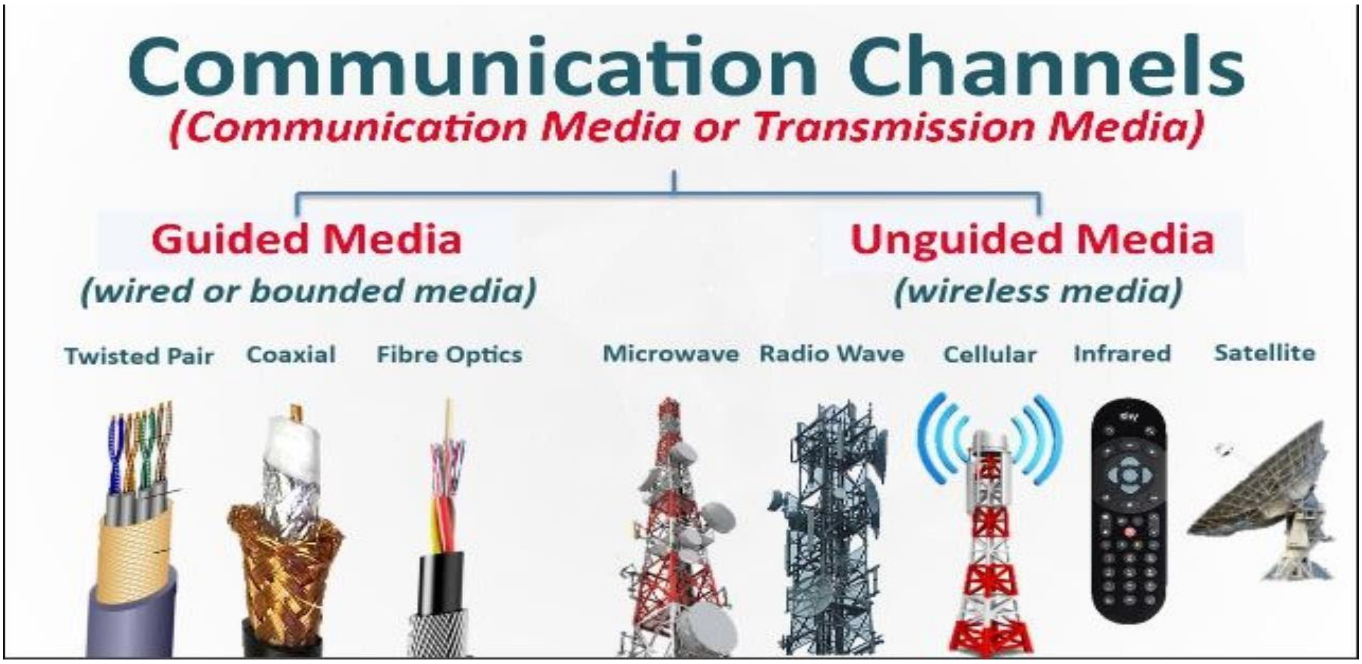
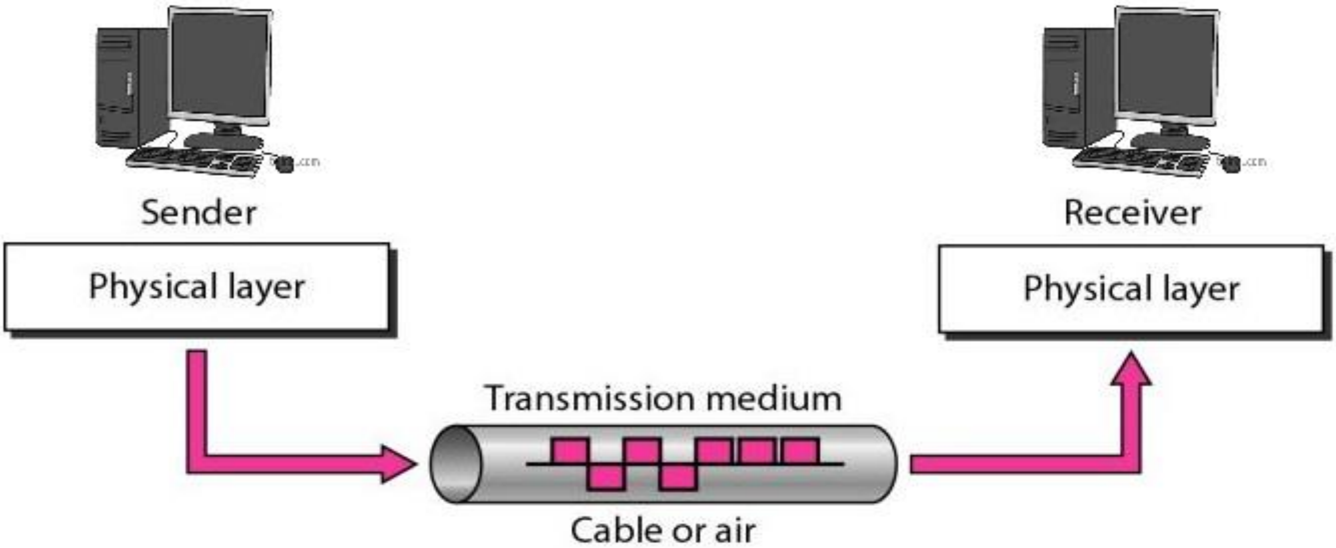
Computer Networking

Data Communication

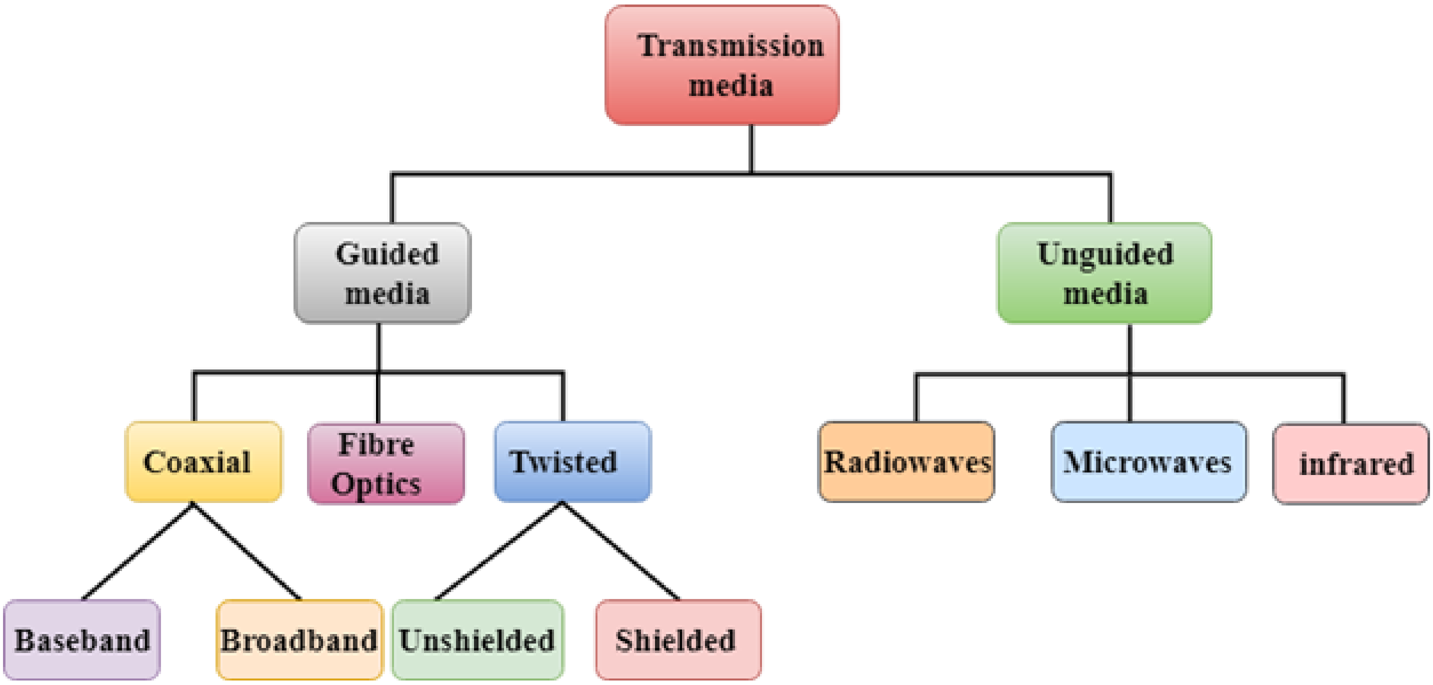
❖ Communication Media / Transmission Media / Communication Line

- The transmission medium is the means by which data/signal is transmitted/carried from one location to another location.
- A part of physical layer in networking / Internet.
- A transmission medium can be defined as anything that can carry analog or digital information from a source to a destination.
- Any message can be sent in digital form as data that can be translated into binary digits(0,1).The signal is encoded/encrypted with these digits. So,data is sent to a network, the network divides into packets, and the packets are formed by the network.
- Just like A mail carrier truck or train/aeroplane could be the delivery medium for a written letter/goods. Similarly,data transfer one nodes to another nodes.





❖ Types of Transmission Media



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1. Guided Media / Bounded Media

- In guided media, data signals flow through the help of wires / cables.
- Data is transmitted through these wires to a particular patch(पथ/रास्ता).
- These wires are made of copper,tin or silver.
- Guided media also known as Bounded media, which are those that provide a conduit (नलिका) from one device to another device.

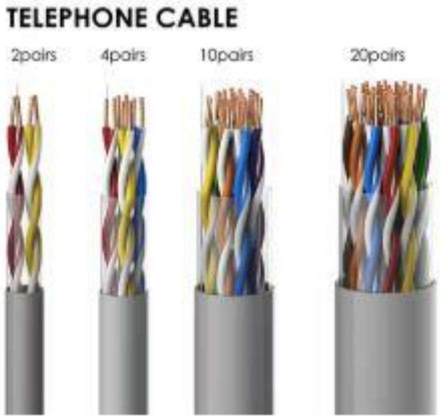
➢ **Bounded transmission media / Guided Media are three types**

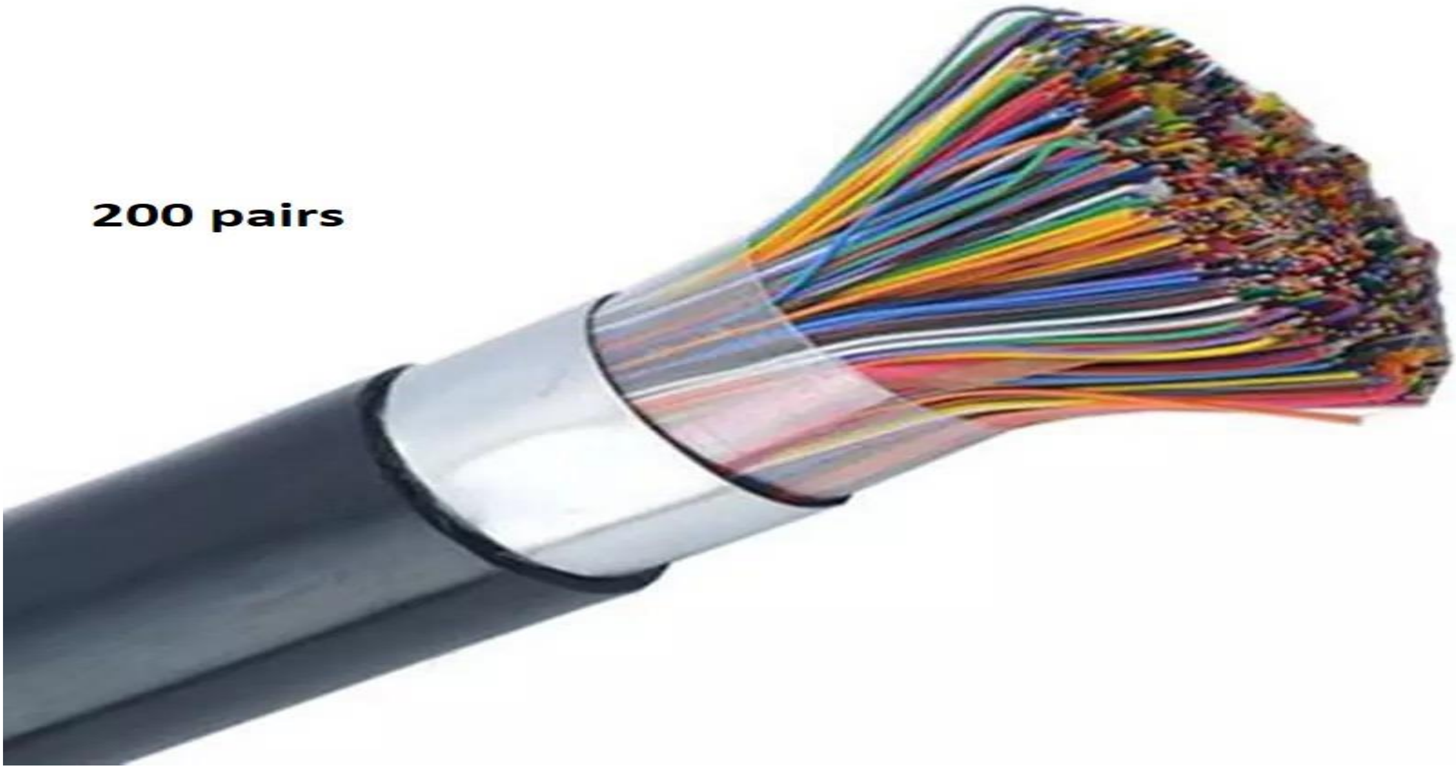
A. Twisted-pair cable
B. Coaxial cable
C. Fiber-optic cable

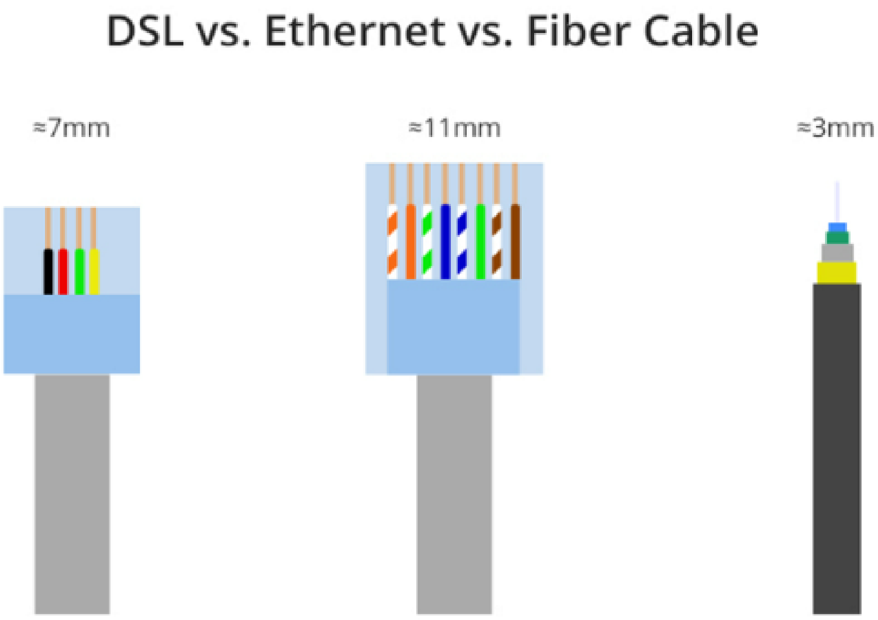
A. Twisted Pair Cable/wire

- The wires are twisted (मुड़ा) together in pairs(जोड़ा).
- The less expensive and most widely used guided transmission medium.
- Twisted pair (TP) maybe used to transmit both analog and digital signal.
- It is lightweight, cheap, can be installed easily, and they support many different types of network.
- Over longer distances, cables may contain hundreds of pairs.
- When data are transferred through this cable, the signal becomes weak after some distance. To maintain the signal, amplifiers are used for analog signals every 5 to 6 km, while repeaters are used for digital signals every 2 to 3 km.
- An amplifier increases the strength of a signal, whereas a repeater regenerates the signal before forwarding it.
- Example of Twisted pair cable : Ethernet cable (CAT5e/CAT6 / CAT6A) , telephone cable (CAT3),etc… (Here CAT means = Category)
- CAT5e,CAT6,…. Wire used for – Ethernet/LAN networking, Telephone line, IP camera, VoIP phone, PoE (Power over ethernet)

- DSL is not a type of cable; it is a technology in which the same telephone cable is used for the Internet.







DSL = Digital subscriber line
use : Internet over telephone line

Figure: DSL vs Ethernet cable vs fiber cable

Practical

Image link 1 (repair cable)
https://images.tribuneindia.com/cms/gall_content/2018/9/2018_9$largeimg15_Saturday_2018_005100203.jpg

Image link 2 (1000 pairs) : https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRAL_g7tVqqBeviSq0GYOqntKkjqUCcRJkRz-kiOKJiUg&s

Image link 3 (old telephone distribution) :
https://live.staticflickr.com/5250/5251322429_c849ce1b55_b.jpg

Application/Use

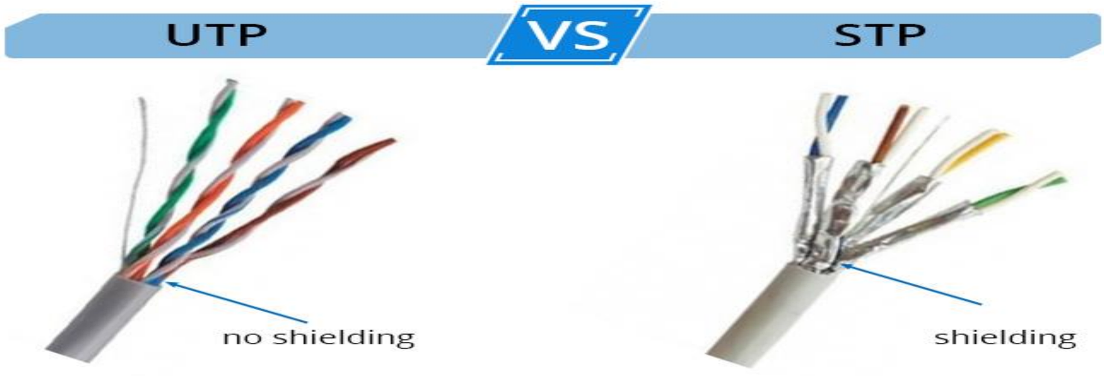
- ✓ It is mainly used in short distance communication like LANs and phone lines.
- ✓ It is the most commonly used medium of telephone / land line.
- ✓ In the telephone system individual residential telephone sets are connected to the local telephone exchange or to end office by twisted pair wire.

✓ It is also commonly used within a building for LAN(Local area network) / Ethernet.

➢ Two types of Twisted pair cable

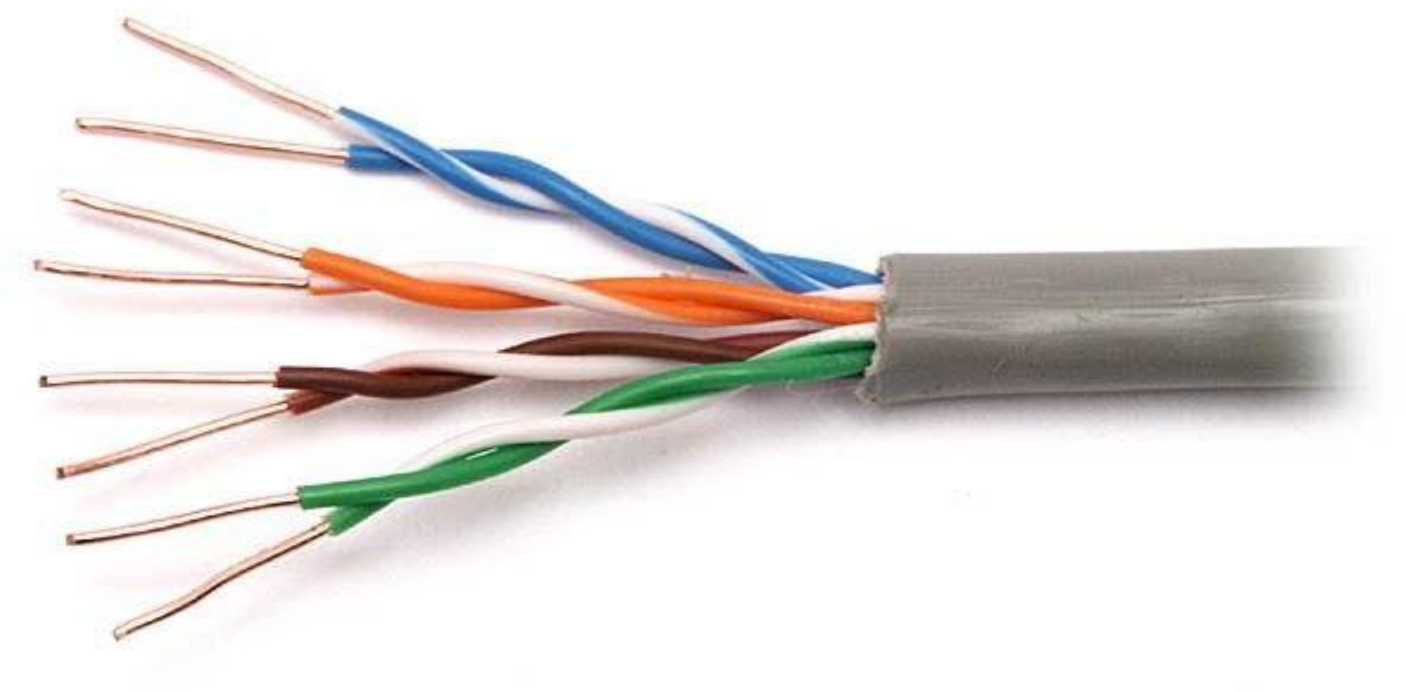
i) Unshielded twisted pair (UTP)

ii) Shielded twisted pair (STP)



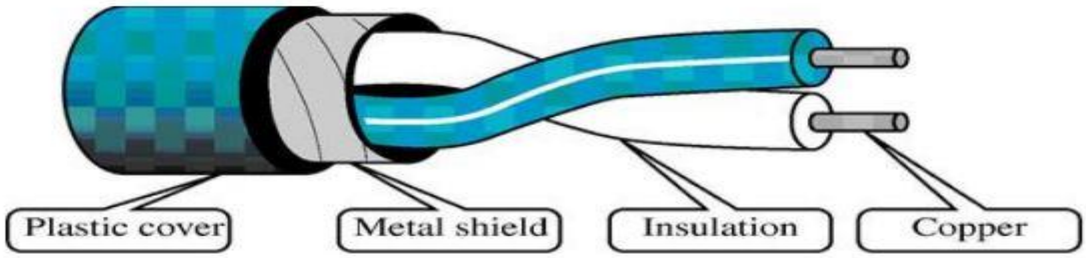
i) UTP (Unshielded twisted pair)

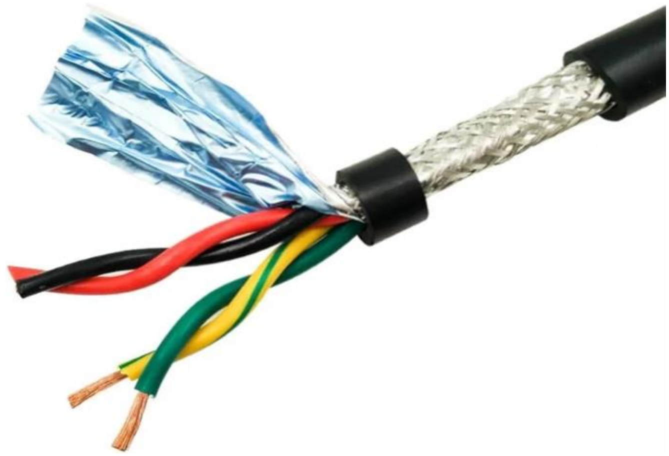
- Usually consists of two copper wires wrapped in individual plastic insulation.
- UTP cables are the most common telecommunications medium.
- The frequency range of the twisted pair cables enable both voice and data transmission.
- UTP cables consist of 2 or 4 pairs of twisted cable. Cable with 2 pair use RJ11 (Telephone) connector and 4 pair cable useRJ-45(PC/Laptop/router,…) connector.



ii) STP (Shielded twisted pair)

- The only difference between STP and UTP is that STP cables have a shielding in usually of aluminum or polyester material between the outer jacket and wire.
- The metal mesh around the insulated wires eliminates crosstalk. Crosstalk occurs when one line picks up some of the other signals traveling down another line.





➢ Difference B/W UTP & STP

Parameters	UTP	STP
Full Form	Unshielded Twisted Pair	Shielded Twisted Pair
Structure	cable with wires that are twisted together.	Twisted pair cable enclosed in foil / shield.
Cost	Cheaper than STP	Costlier than UTP
Weight	Lighter than STP	Heavier than UTP
Noise & interference	Prone to Noise and interference nature	Less prone to noise and interference
Data Speed	Supports slower speed than on STP	Support higher speed than UTP
Grounding of cable	Not required	Required
Target deployments	Locations less prone to interference like offices and homes.	Locations prone to interference like factories and airports

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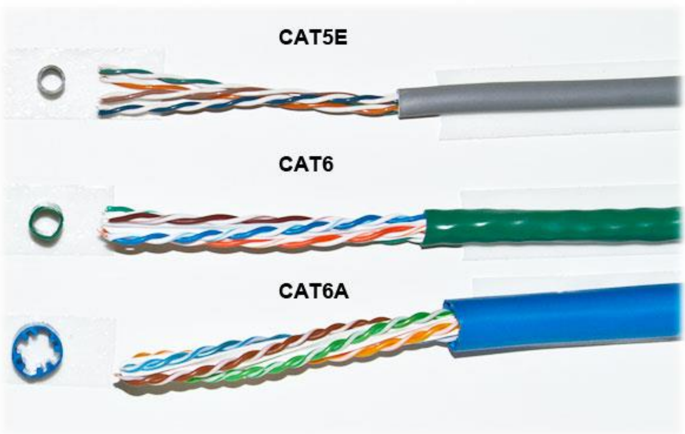
Extra Topic (Not in Syllabus)

Twisted Pair cable			
Cable name	**Connector name**	**use**	**Speed**
CAT3	RJ 11	Telephone line	
CAT5	RJ45	Ethernet LAN, Internet	10-100mbps
CAT5E	RJ45	Ethernet LAN, Internet	1GBps
CAT6 / CAT6A	RJ45	Ethernet LAN, Internet	1GB – 10GBps
CAT6A	RJ45	Ethernet LAN, Internet	10GBps

RJ45

RJ11

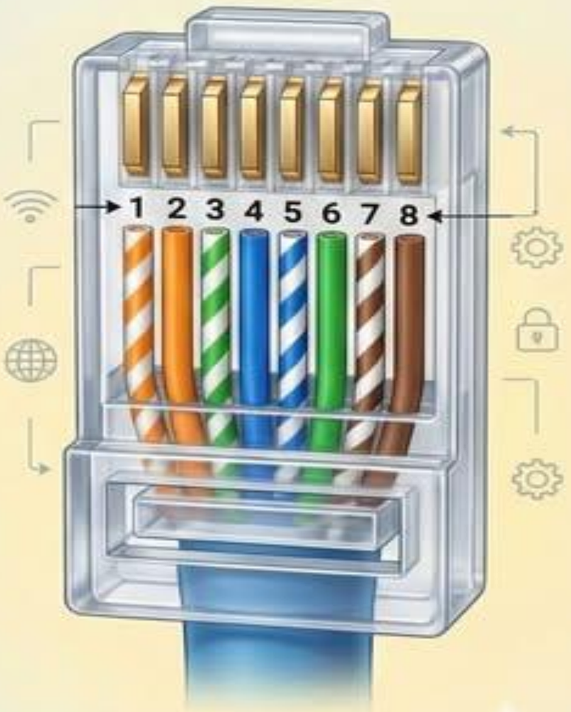


RJ45 LAN Cable Wiring Color Code (T568A & T568B)

T568A Standard

T568A Color Code	
Pin 1	White-Green
Pin 2	Green
Pin 3	White-Orange
Pin 4	Blue
Pin 5	White-Blue
Pin 6	Orange
Pin 7	White-Brown
Pin 8	Brown

T568B Standard



T568B Color Code	
Pin 1	White-Orange
Pin 2	Orange
Pin 3	White-Green
Pin 4	Blue
Pin 5	White-Blue
Pin 6	Green
Pin 7	White-Brown
Pin 8	Brown

One Wrong Wire = No Internet

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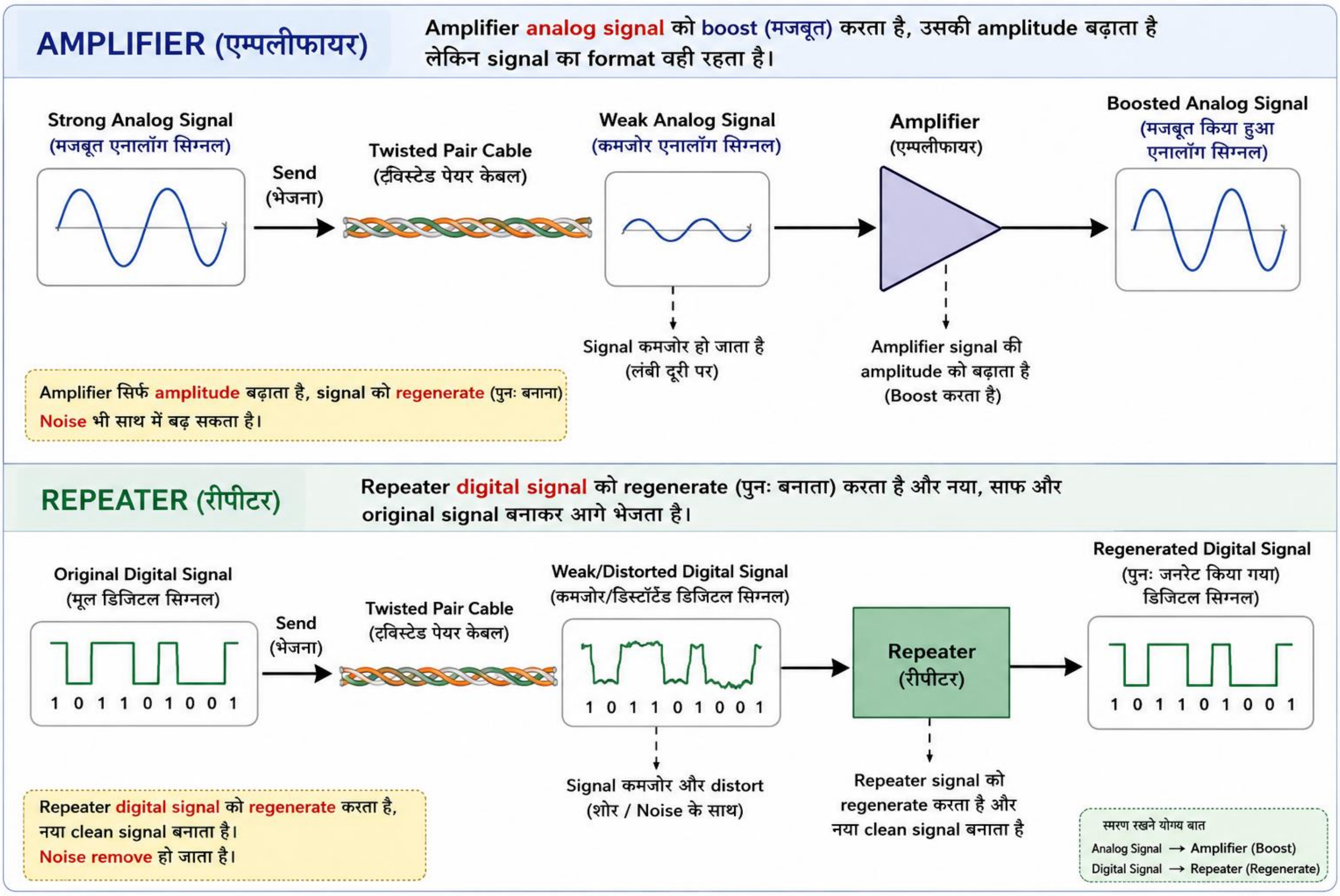

AMPLIFIER (एम्पलीफायर)
Amplifier analog signal को boost (मजबूत) करता है, उसकी amplitude बढ़ाता है
लेकिन signal का format वही रहता है।
Strong Analog Signal
(मजबूत एनालॉग सिग्नल)
Send
(भेजना)
Twisted Pair Cable
(ट्विस्टेड पेयर केबल)
Weak Analog Signal
(कमजोर एनालॉग सिग्नल)
Amplifier
(एम्पलीफायर)
Boosted Analog Signal
(मजबूत किया हुआ
एनालॉग सिग्नल)
Signal कमजोर हो जाता है
(लंबी दूरी पर)
Amplifier signal की
amplitude को बढ़ाता है
(Boost करता है)
Amplifier सिर्फ amplitude बढ़ाता है, signal को regenerate (पुनः बनाना)
Noise भी साथ में बढ़ सकता है।
REPEATER (रीपीटर)
Repeater digital signal को regenerate (पुनः बनाता) करता है और नया, साफ और
original signal बनाकर आगे भेजता है।
Original Digital Signal
(मूल डिजिटल सिग्नल)
1 0 1 1 0 1 0 0 1
Send
(भेजना)
Twisted Pair Cable
(ट्विस्टेड पेयर केबल)
Weak/Distorted Digital Signal
(कमजोर/डिस्टॉर्टेड डिजिटल सिग्नल)
1 0 1 1 0 1 0 0 1
Repeater
(रीपीटर)
Regenerated Digital Signal
(पुनः जनरेट किया गया)
डिजिटल सिग्नल)
1 0 1 1 0 1 0 0 1
Signal कमजोर और distort
(शोर / Noise के साथ)
Repeater signal को
regenerate करता है और
नया clean signal बनाता है
Repeater digital signal को regenerate करता है,
नया clean signal बनाता है।
Noise remove हो जाता है।
स्मरण रखने योग्य बात
Analog Signal → Amplifier (Boost)
Digital Signal → Repeater (Regenerate)

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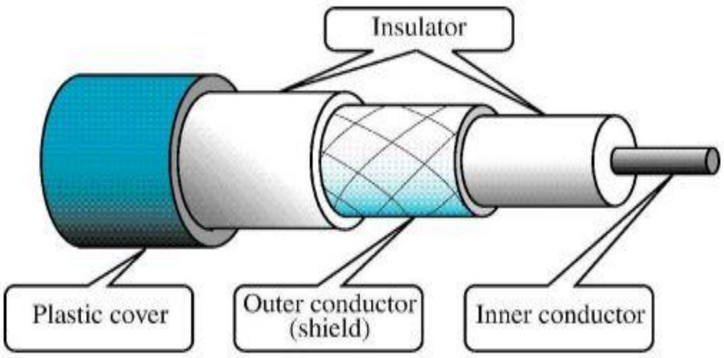
B. Co-axial cable

- The name "coaxial" comes from the fact that it contains two conductors that share a common axis.
- The inner conductor is made of copper and is surrounded by PVC / Teflon insulation.
- The outer conductor is made of metal foil(पति),, a metal mesh (braid) (जाली), or both.
- The outer metallic conductor is used as a shield against noise and electromagnetic interference (EMI).
- The outermost part is a plastic cover that protects the entire cable.
- Coaxial cable is much less susceptible(अतिसंवेदनशील) to interference and crosstalk than twisted-pair cable.
- Coaxial cable is used to transmit both analog and digital signals.



➢ Application /Use

✓ TV distribution cable /Cable TV – Present time

✓ Long distance telephone transmission – Not at the present time

✓ Short distance communication links (pc to pc) – Not at the present time

✓ LAN - Not at the present time

✓ CCTV – Present time

✓ Cable modem internet - present time

➢ Co-axial cables usually for data transmission are of two types,its called Ethernet network standards.

Standard	Cable used	Nickname	Max length

10BASE5	Thick Coaxial cable (RG-8)	Thicknet	500m
10BASE2	Thin coaxial cable (RG-58)	Thinnet	200m

here

10 = 10 Mbps speed

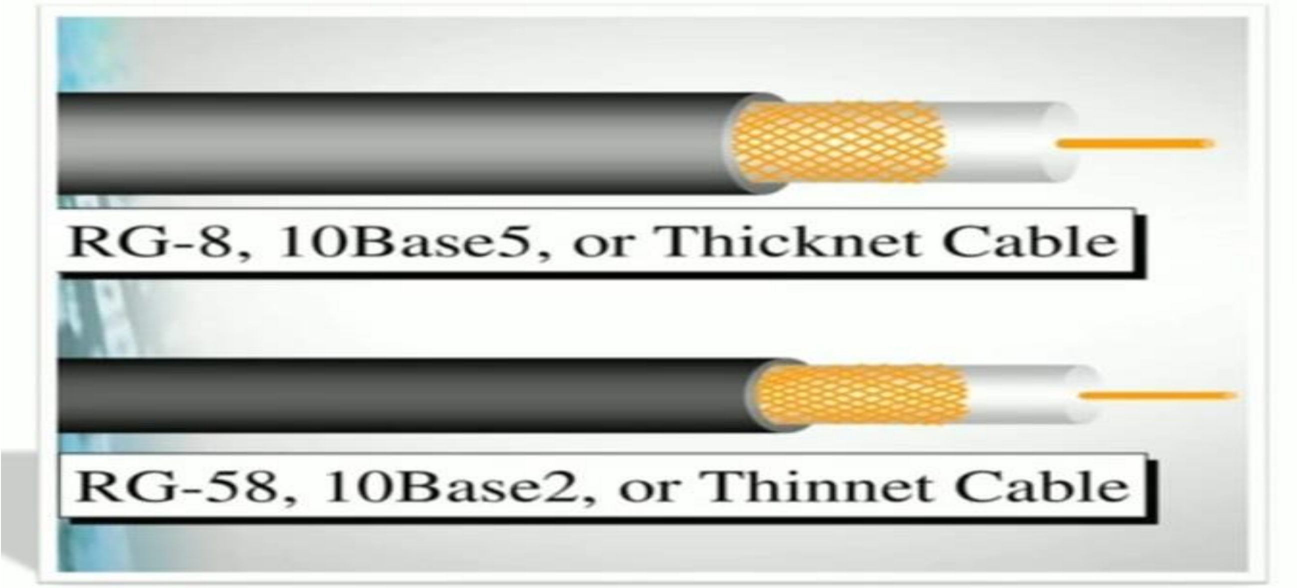
BASE = Baseband transmission (Digital signal is transmitted directly without modulation)

5 = Approx. 500m for reliable data transmission

2 = Approx. 200m for reliable data transmission

What is Modulation : The process of combining a message signal with a carrier signal for transmission.

What is carrier signal : A high frequency signal used to carry information form one place to another.



➢ BaseBand

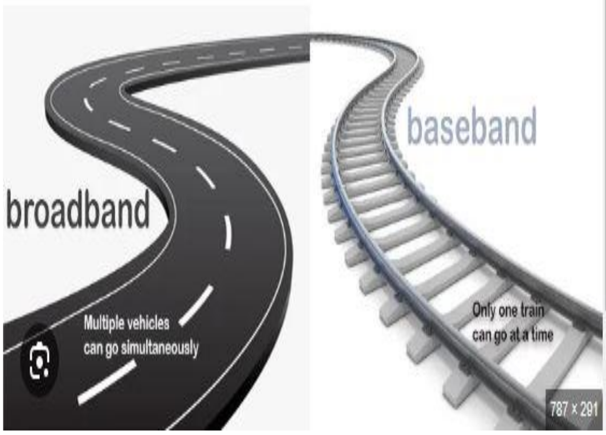
- It is mostly used for LANs (local area network).
- Baseband transmits only one signal at a time at high speed.
- In traditional baseband coaxial cable, a repeater is required approximately every 1000 feet (300 meters) to regenerate the digital signal.
- No modulation
- Use : Ethernet LAN

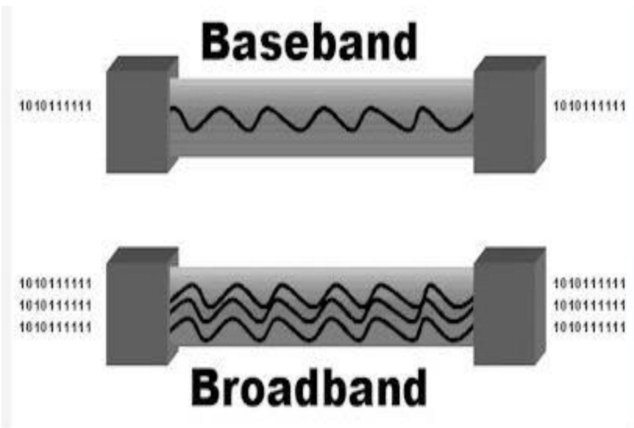
➢ BroadBand

- It uses analog transmission over standard cable television (CATV) cables.
- It transmits multiple signals simultaneously using different frequencies.
- It covers a larger area than baseband coaxial cable.
- Modulation required
- Use : TV, Radio, DSL(digital subscriber line), Mobile network

Quick Difference

Baseband	Broadband
Digital transmission	Analog transmission
One signal at a time	Multiple signals at the same time
Mostly used in LAN	Used in Cable TV and Cable Internet
Uses repeaters for long distances	Uses amplifiers for analog signals





➢ Bandwith

- The Bandwidth is the maximum amount of data that can be transmitted over a network connection in a given amount of time.
- In other words, bandwidth is the maximum data transfer capacity of a communication channel.
- Bandwidth is usually measured in Mbps (Megabits per second) or Gbps (Gigabits per second).

➢ Throughput

- Throughput is the actual amount of data successfully transferred from one end to another in a given amount of time.
- Example: If the bandwidth is 30 Mbps, but only 25 Mbps is actually received, then the throughput is 25 Mbps.

➢ Latency

- Latency is the time taken for data to travel from the source to the receiver.
- It depends on factors such as network traffic, transmission medium, and distance.
- Latency is measured in milliseconds (ms).

All Example :

Bandwidth: 100 Mbps (maximum capacity) [data 100mpbs के स्पीड से भेज सकता हैं }

Throughput: 80 Mbps (actual speed achieved) {data 80 mbps पहुँचा }

Latency: 20 ms (time taken for data to reach the destination) {समय 20मिलीसेकंड लिया}

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Extra Topic (Not in Syllabus)

Twisted Pair cable		
Cable name	**Connector name**	**use**

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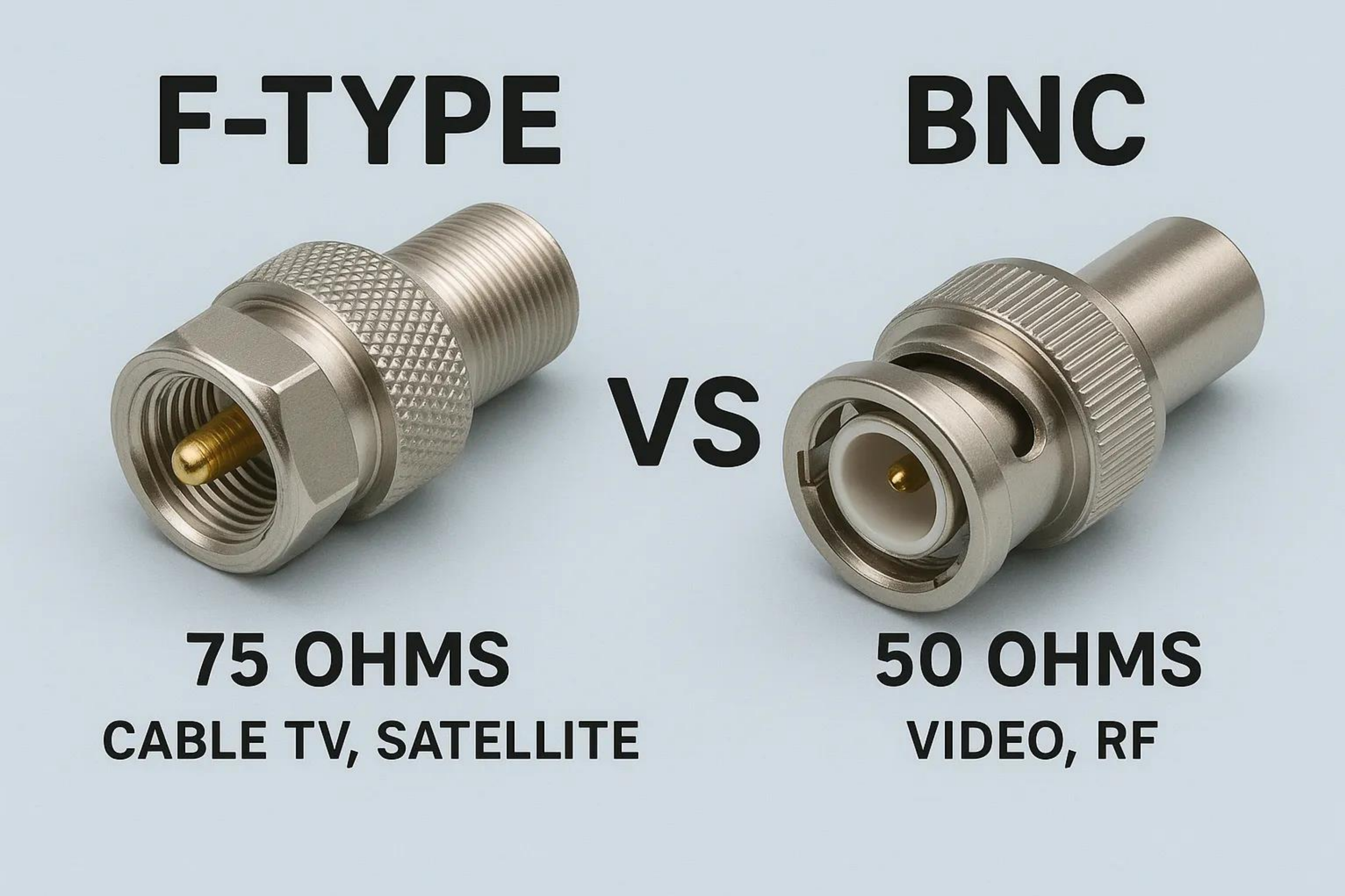
RG-6	F-type connector	Cable tv, cable broadand internet
RG-59	BNC connector	Analog CCTV camera
RG-58	BNC connector	Old ethernet LAN (10BASE2), radio
RG-11	F-type connector	Long-distance cable tv/ broadband



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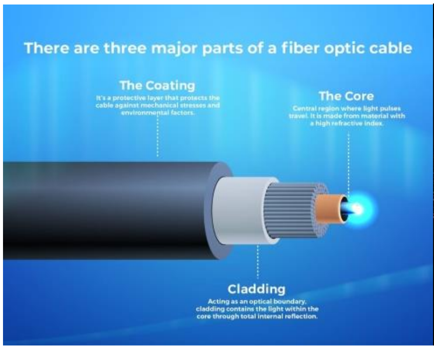
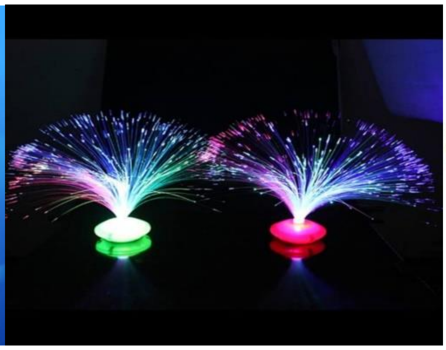
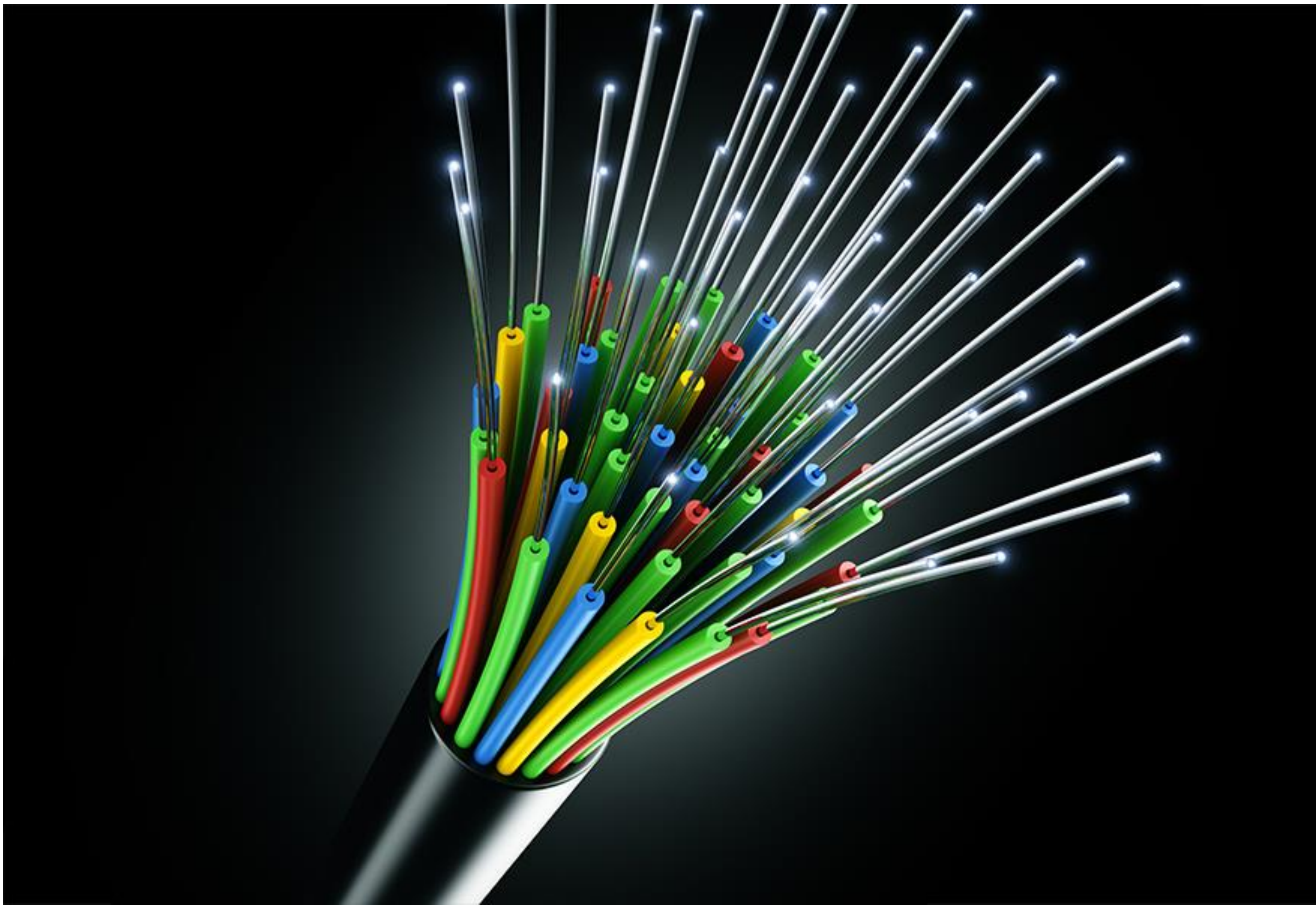
C. Fiber-optic cable OR Optical Fiber

- A fiber-optic cable is made of glass or plastic and transmits signals in the form of light.
- It works on the principle of **Total internal reflection** (TIR)
- A light pulse represents a binary 1, while the absence of a light pulse represents a binary 0.
- The bandwidth of an optical transmission system is very high.
- It supports extremely high-speed data transmission.
- The transmission medium is an ultra-thin fiber made of glass or plastic.

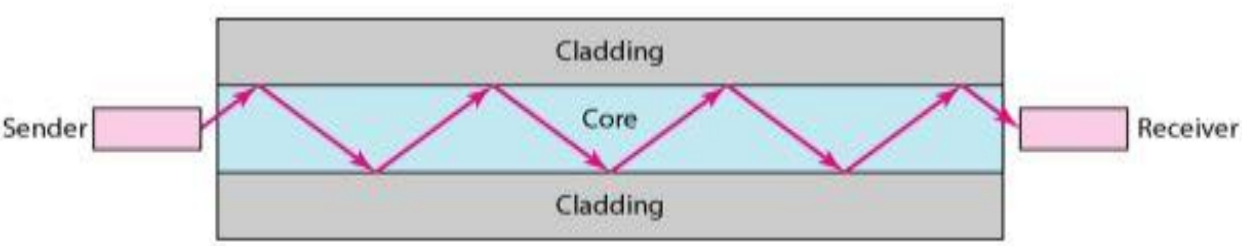






An optical fiber has a cylindrical shape and consists of three concentric sections:

(i) Core **(ii) Cladding** **(iii) Jacket**



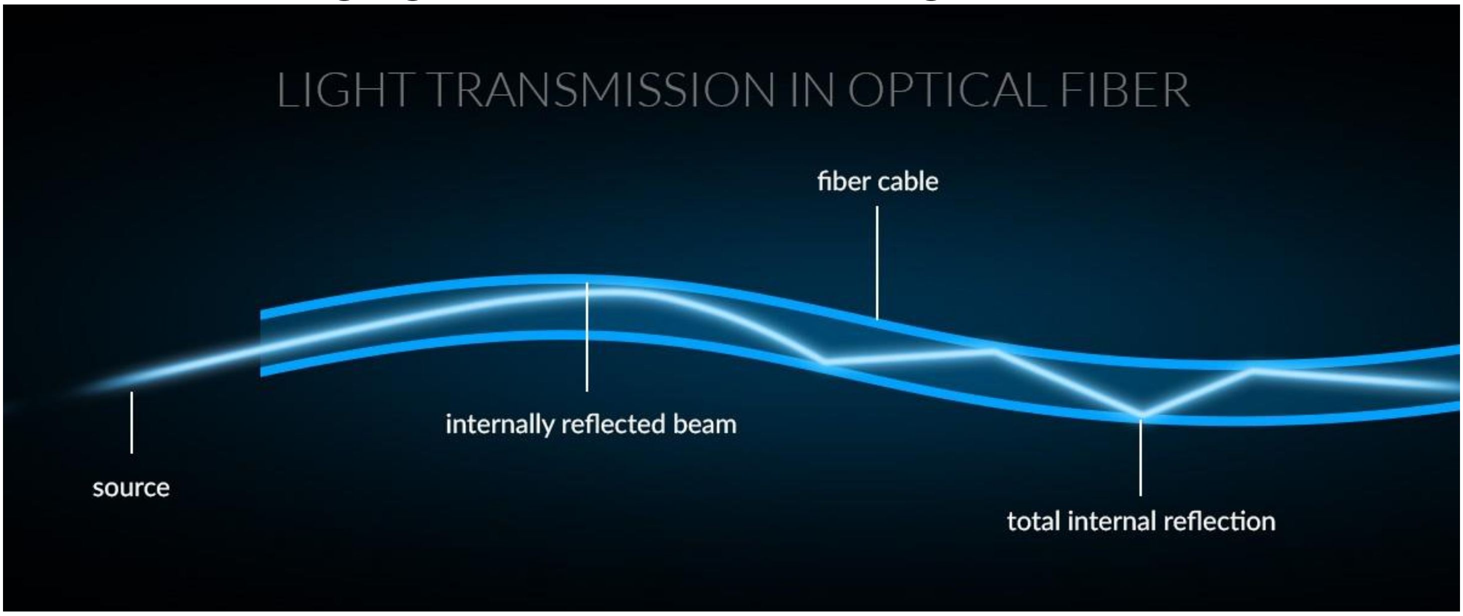
1. **Core:-** It's the inner most section is made of glass or plastic and is surrounded by its own cladding. The core diameter is in the range of 8 to 50 µm.

2. **Cladding: -** A glass or plastic coating that has optical properties different from those of the core having a diameter of 125 µm. The cladding acts as a reflector to light that would otherwise escape the core.

3. **Jacket: -** The outer most layers surrounding caddied fiber is the jacket. Jacket is composed of plastic or other material layer to protect against moisture, cut, crushing and other environmental dangers.

➢ Optical Fiber Communication

- A transmitter (light source) at the sender's end sends light through the optical fiber.
- A receiver at the other end uses a light-sensitive device to detect the presence or absence of light, indicating binary 1 or 0.
- **Total Internal Reflection (TIR)** is the complete reflection of light back into a denser medium when it travels from a denser medium to a rarer medium at an angle greater than the critical angle.



➢ Two types of light sources are commonly used in fiber-optic communication systems:

- The Light Emitting Diode (LED)
- Injection Laser Diode (ILD).

➢ Application/Use

✓ Telephones

✓ including cellular wireless Internet

- ✓ LANs - local area networks, WAN
- ✓ CCTV Camera
- ✓ Medical Application
- ✓ Transportation – smart lights and highways
- ✓ Military

Practical

Video 1 = Fiber stripping (फाइबर के बाहरी पार्ट को हटाना)

https://www.youtube.com/shorts/DrIaMo7MuLk

Video 2 =2 fiber cable join process

https://www.youtube.com/shorts/-Fm7qwMklh0 (outside machine)

https://www.youtube.com/shorts/YWqR3Ae3AZ4 (inside machine)

Video 3 = Inside optical fiber

https://www.youtube.com/shorts/-xf1BY1L6r8

Video 4 = Check fiber cable fault / damage

https://www.youtube.com/shorts/D7BAKs4Xq4w

https://www.youtube.com/shorts/KCunTO9-5iA

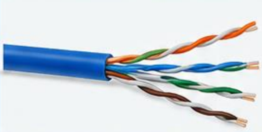
Video 5 = TIR video of fiber cable

https://www.youtube.com/shorts/PhJ8uxlOqO4

➢ Difference / Characteristics of cable media:-

Factor	UTP	STP	Co-axial	Fiber optics
Cost	Low	Moderate	Moderate	Highest
Installation	Easy	Fairly easy	Fairly easy	Difficult
Data rate	1 to 155 mbps	1to 155 mbps	500 mbps	2 GBPS
Node capacity	2	2	30-100	2
Attenuation	High(100's of meter)	High(100's of meter)	Lower (range of few km's)	Lowest (10 Km's)
EMI	Most vulnerable	Less vulnerable than UTP	Less vulnerable than UTP	Not effected by EMI
Bandwidth	Low	Moderate	Moderatly high	Very high
Signals	Electrical	Electrical	Electrical	Light

DIFFERENCE BETWEEN UTP, STP, COAXIAL AND FIBER OPTIC CABLE

Feature	UTP CABLE (Unshielded Twisted Pair)	STP CABLE (Shielded Twisted Pair)	COAXIAL CABLE	FIBER OPTIC CABLE
Cable Image				
Construction	4 Pairs of copper wires twisted together	Twisted pairs of copper wires with overall shield (foil/braid)	Central copper conductor, dielectric insulation, metal shield, outer jacket	Glass or plastic core, cladding, protective coating, outer jacket
Shielding	No Shielding	Shielded (Foil or Braid)	Shielded (Metal Braid/Shield)	No Shielding (Uses light)
Medium / Signal	Electrical (Copper)	Electrical (Copper)	Electrical (Copper)	Light (Optical Fiber)
Data Transmission	Digital	Digital	Analog / Digital	Digital (Light Pulses)
Speed	Up to 1 Gbps (10 Gbps for Cat 6a/7)	Up to 1 Gbps (10 Gbps for Cat 6a/7)	Up to 100 Mbps – 1 Gbps (Depends on type)	Very High Speed (Mbps to Tbps)
Distance	Up to 100 meters	Up to 100 meters	Up to 500 meters (Depends on type)	Up to 2 km to 100+ km (Depends on type)
Interference (Immunity)	Low (More Affected by EMI)	High (Better protection from EMI)	High (Good protection from EMI)	Very High (Immune to EMI and RFI)
Common Uses	LAN, Telephone, Networking	LAN, Industrial Networks, High EMI areas	Cable TV, Internet (Broadband), CCTV	Long Distance Communication, Internet Backbone, Data Centers
Connector	RJ45	RJ45	BNC, F-Type, N-Type	SC, LC, ST, FC
Cost	Low	Medium	Medium	High
SUMMARY	UTP is low cost, easy to install and used in LANs, but more susceptible to interference.	STP has shielding, better performance in noisy environments, used in LANs and industries.	Coaxial cable is durable, has good shielding and used in cable TV and broadband connections.	Fiber optic cable offers very high speed and long distance with no interference, used in advanced communication systems.

=====================HIRA KUMAR=======================